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Co-Deposited Inverted Perovskite Photovoltaics towards 27% Efficiency via Vertical Redistribution of Self-Assembled-Molecules and In-Situ Crosslinking

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Abstract

Co-deposited perovskite solar cells simplify the fabrication process, yet excessive aggregation of self-assembled molecules (SAMs) during crystallization leads to poor interfacial adhesion, thereby limiting performance and stability. This study designs an asymmetric SAM (PhBr-4PACz) with steric hindrance to mitigate self-aggregation by suppressing the face-to-face stacking of planar conjugated cores, thereby enriching the presence of co-deposited SAMs at the bottom interface with improved interfacial adhesion and coverage. Its deepened energy level by bromine group and observed p-type doping effect further promotes hole extraction. To prevent SAM diffusion under thermal stress, 1-Allyl-3-vinylimidazolium chloride was introduced to seal grain boundaries by in-situ crosslinking to suppress the upward diffusion of SAMs and release stress. Here we show that the optimized inverted devices achieve a certified power conversion efficiency of 27.03% and retain over 90% of their initial efficiency after 2,000 hours of continuous illumination at the maximum power point (85°C).

Introduction

To date, the certified power conversion efficiency (PCE) of inverted perovskite solar cells (IPSCs) has surpassed 27.0%, primarily driven by the development of self-assembled molecules (SAMs).[1-6] Recent reports indicate that when added to the perovskite precursor solution, SAMs spontaneously deposit onto the substrate to form hole selective layer (HSL).[7,8] Compared to conventional sequential deposition, this co-deposition technique eliminates the pre-deposition step of HSL, thereby effectively reducing the production cost.[7,9] However, the PCE of co-deposited IPSCs still lags behind that of sequentially deposited IPSCs.[8,10] A critical barrier hindering further breakthroughs in co-deposited devices is the poor adhesion of SAM at the bottom interface caused by self-aggregation and diffusion of SAM, leading to excessive accumulation of SAM at the top surface and in the bulk phase of the perovskite layer.[7,8,10,11]

During the co-deposition process, the self-aggregation tendency of the SAMs reduces their solubility, so that SAMs tend to precipitate on the top surface of perovskite during the early stages of crystallization of the perovskite precursor. Subsequently, the enrichment of SAMs at the top surface can lead to energy-level mismatch, resulting in increased recombination and energy loss at the interface. Zhu et al. found that co-deposited SAM tends to accumulate at the top surface and suggested employing IPA cleaning as a means to mitigate this issue. However, they also observed that a substantial amount of SAM may remain in the bulk phase, which could directly lead to a decrease in long-term stability.[7] He et al. further highlighted that SAMs exhibit excessive aggregation and jamming at grain boundaries (GBs), inducing severe recombination centers.[8] Moreover, the carbazole or triphenylamine moieties in SAMs exhibit strong electron-withdrawing characteristics, imparting a discernible p-type doping effect on the perovskite.[11,12] More critically, when SAMs are rather accumulated on top surface and in the bulk of perovskite layer, that means that there are insufficient SAMs located at the bottom interface. This outcome leads the poor adhesion between perovskite layer and ITO (or FTO) substrate. The poor adhesion at the buried interface also induces severe defect states and energy level misalignment.[13-16] Therefore, optimizing the vertical distribution profile of SAMs throughout the absorption layer is essential to improve the PCE of co-deposited devices beyond 27%.[7,10]

While the self-aggregation behavior determines the vertical profile of SAMs in co-deposited IPSCs, the design of SAMs and their capability to resist self-aggregation is still unclear. In an attempt to understand self-aggregation, our group has previously developed a SAM, namely Ph-4PACz, in which symmetrically bonded, large-dihedral-angle phenyl rings are used as terminal groups to disrupt the planarity of the conjugated core of SAMs and suppress aggregation.[17,18] Recently, Xue et al. developed SAX, which features orthogonal π scaffolds by introducing orthogonally coupled units into the carbazole unit, to reduce molecular packing and form disordered structures.[19,20] Additionally, asymmetric structures, such as 4-PhCz, DMICPA, and 4PABCz, have also shown potential in reducing self-aggregation and improving the adhesion of SAMs at the bottom interface.[4,21,22] However, the issue of self-aggregation of SAMs in co-deposited IPSCs is rarely studied.[23,24] A design strategy of SAMs dedicated to reduce their self-aggregation in the perovskite precursor solution and thereby change the vertical distribution profile of SAMs is urgently needed for co-deposited IPSCs.

Due to their small molecular size, SAMs tend to desorb under thermal stress and diffuse into the perovskite. Since co-deposition process potentially hinders the distribution and adsorption of SAMs at the bottom interface, the diffusion of SAMs in co-deposited PSCs is exacerbated under thermal stress.[25,26] In order for SAMs to diffuse upwards, the GBs in the perovskite absorber provide the pathway for molecular movement. Consequently, sealing or crosslinking the GBs is essential.[27] Among established crosslinking strategies, epoxy heterocycles, disulfide bonds, alkenes, and alkynes are capable of in situ polymerization, as exemplified by COETC, TA-NI, SBMA, and PAI.[28–31] In particular, alkene-based crosslinkers have shown enhanced reactivity at low temperatures. For example, imidazolium-based ionic liquids containing alkenes can be crosslinked at temperatures below 100°C, making them promising candidates for perovskite application.[32-34] Overall, the implementation of GB crosslinking in co-deposited architectures is also urgently needed to effectively mitigate the stability degradation arising from SAM diffusion along GBs.

In this work, an asymmetric design of SAM, namely PhBr-4PACz, with bulky steric hindrances, was proposed and synthesized by functionalizing the carbazole unit with a phenyl substituent on one end and a bromine substituent on the other end. The large dihedral angle of the phenyl group disrupts the face-to-face stacking of the planar conjugated core, effectively suppressing the self-aggregation of PhBr-4PACz in perovskite precursor. Compared to Me-4PACz, PhBr-4PACz shows a lower enrichment at the top surface and in bulk-phase of the perovskite layer and a higher enrichment at the bottom interface for the co-deposition process. Consequently, the interfacial adhesion at bottom interface is greatly improved due to the redistribution of SAMs vertically. The bromine substituent of PhBr-4PACz further deepens the highest occupied molecular orbital (HOMO) energy level of SAMs and induces a stronger p-type doping effect for the bottom region of perovskite phase, resulting in enlarged quasi-fermi level splitting (QFLS). As a result, the extraction of hot and cold holes at the buried interface was enhanced, and non-radiative recombination losses were significantly reduced. However, significant diffusion of SAMs at the bottom interface along GBs remains inevitable during aging. To address this, the introduction of highly fluid, and cross-linkable (AVIMCl) molecules on the perovskite surface is proposed. AVIMCl can penetrate along the GBs to the bottom interface and crosslink at low temperature to not only seal the GBs but also release the residual stress of perovskite layer. As a result, the optimized inverted device achieved a champion PCE of 27.26%, certified at 27.03% (quasi-steady-state 26.50%), setting a record for IPSCs without pre-deposited HTL (also known as co-deposited PSCs) to date. Importantly, the unencapsulated devices maintain 96.6% and over 90% of the initial efficiency after 2000 hours of maximum power output (MPP) tracking at 65℃ under ISOS-L-2 protocol and 2000 hours of heating at 85°C, respectively, confirming excellent operational and thermal stability.

Results

Properties, Distribution, and Crystallization Regulation of PhBr-4PACz

The molecular structures and electrostatic potentials (ESP) of Me-4PACz and PhBr-4PACz are depicted in Fig. 1a. PhBr-4PACz was designed by attaching a bromine atom and a phenyl group to opposite sides of the carbazole unit in 4PACz. The detailed synthesis of PhBr-4PACz molecules is depicted in the Supplementary Fig. 1, and all the synthesized key intermediates and the final compounds were characterized by nuclear magnetic resonance (NMR) and high-resolution mass spectrometry to examine their molecular structures and purity (Supplementary Figs. 2-13). Due to the introduction of the strongly electron-withdrawing bromine group and the conjugated phenyl group, PhBr-4PACz exhibits an enhanced dipole moment and extended highest occupied molecular orbital (HOMO) distribution (Fig. 1b). Density functional theory (DFT) calculations show that, compared to Me-4PACz, the dipole moment of PhBr-4PACz significantly increases from 1.68 D to 3.76 D, and the HOMO energy level deepens from -5.16 eV to -5.41 eV, effectively modulating the work function and enhancing charge transport. Furthermore, the lowest unoccupied molecular orbital (LUMO) energy levels of PhBr-4PACz and Me-4PACz are -1.03 eV and -0.61 eV, respectively, both of which are considerably below the conduction band minimum (CBM) of perovskite, effectively blocking electron injection (Supplementary Fig. 14). The rigid steric hindrance and halogen bonding interactions imparted by the phenyl and halogen groups increase the lattice energy and suppress thermal decomposition. Thermogravimetric analysis (TGA) reveals that the temperatures at which a 5% weight loss occurs for Me-4PACz and PhBr-4PACz are 319°C and 404°C, respectively, indicating that PhBr-4PACz exhibits superior thermal stability, making it suitable for devices (Supplementary Fig. 15). In contrast to the planar conjugated core in Me-4PACz, PhBr-4PACz exhibits a significant dihedral angle of 38.3° between its phenyl substituent and carbazole core (Supplementary Fig. 16). The incorporation of the bulky phenyl group introduces substantial steric hindrance, which is likely to affect molecular stacking and aggregation behavior.

To gain further insight into how the bulky configuration of PhBr-4PACz modulates SAM aggregation, the single-crystal structures of the corresponding conjugated units were analyzed.[35,36] Specifically, the single-crystal structure of the conjugated unit in Me-4PACz (3.6-dimethyl-9*H*-carbazole, Me-Cz) and PhBr-4PACz (3-bromo-9-methyl-6-phenyl-9*H*-carbazole, PhBr-Cz) were synthesized, and the corresponding crystal structures have been deposited in the Cambridge Structural Database under deposition numbers CCDC 2512507 (PhBr-Cz) and CCDC 2512509 (Me-Cz), respectively. Both molecules exhibited parallel (face-to-face) and T-shaped (edge-to-face) stacking characteristics in their crystalline states (Figs. 1c, d). In the T-shaped configuration, the C-H$\cdots\pi$ distances for Me-Cz and PhBr-Cz were 2.65/2.88 Å and 2.55/3.79 Å, respectively, suggesting that Me-Cz has slightly strengthened edge-to-face interactions. However, in the face-to-face stacking, the vertical stacking distances and longitudinal slip distances (the displacement between adjacent molecules along the stacking direction) between Me-Cz and PhBr-Cz were 2.63/5.96 Å and 5.00/9.52 Å, respectively, reflecting a diminished parallel interaction in the PhBr-Cz conjugated core and a significant reduction in the overlap area, thus alleviating the face-to-face stacking behavior. Furthermore, both the trimer and dimer models of the conjugated units showed that the carbazole core of Me-Cz tends to overlap, while the conjugated carbazole core of PhBr-Cz is significantly offset (Supplementary Figs. 17 and 18), which indicates that the introduction of the phenyl group with large dihedral angle disrupts the face-to-face stacking of the carbazole. The binding energy of the PhBr-4PACz dimer, with reduced face-to-face stacking area, is 53 meV greater than that of Me-4PACz (Fig. 1e), confirming that the disruption of the face-to-face stacking diminishes the strength of the van der Waals interactions, resulting in a reduced aggregation tendency for PhBr-4PACz compared to Me-4PACz.

To investigate the effect of SAM self-aggregation on its vertical distribution within co-deposited perovskite (SAM) films, time-of-flight secondary ion mass spectrometry (ToF-SIMS) analyses based on the same SAM content were conducted (Fig. 1f and Supplementary Fig. 19). First, PO_2^- signals were observed at both the FTO/perovskite and perovskite/ETL interfaces in the Me-4PACz and PhBr-4PACz films, indicating that the migrations of the SAM towards upper and bottom interfaces during the crystallization process are common for co-deposited perovskite layer.

Notably, the intensity of PO_2^- at the upper interface and in the bulk phases of the Me-4PACz film was much more pronounced than in PhBr-4PACz. More importantly, the intensity at the bottom was diminished for Me-4PACz film, but remained significant for PhBr-4PACz film. This suggests that a substantial amount of Me-4PACz preferentially accumulates at the top surface and GBs of the perovskite, leading to a lower enrichment at the bottom interface. To assess the impact of SAM's aggregation on crystallization, dynamic light scattering (DLS) was first used to characterize the cluster size distribution in perovskite precursor solutions with the same SAM content. Supplementary Fig. 20 shows that both PhBr-4PACz and Me-4PACz exhibit enlarged micelle sizes relative to the precursor solution without SAM, which can be attributed to the coordination between phosphate groups and the perovskite precursors, consistent with the redshift observed in the P=O and P-O-H groups in the FTIR spectra (Supplementary Fig. 21). However, compared to the uniform micelle size distribution (~390 nm) induced by PhBr-4PACz, Me-4PACz displays a more heterogeneous distribution. In addition to a broad micelle size distribution around 273 nm, a small fraction of larger micelles (~1800 nm) is also observed, showing the excessive aggregation of Me-4PACz in perovskite precursor solution. Upon further analysis, the average micelle size distributions induced by Me-4PACz and PhBr-4PACz were determined to 390 nm and 431 nm, respectively. The crystallization process of the co-deposited films was further monitored using in situ photoluminescence (PL) spectroscopy. During the spin-coating process (Supplementary Fig. 22), after the addition of an anti-solvent, the PL intensity of PhBr-4PACz increased rapidly and continuously compared to Me-4PACz, indicating that PhBr-4PACz facilitated a higher nucleation density. During the annealing stage (Fig. 1g and Supplementary Fig. 23), the PL intensity of perovskite films first increased due to surface crystallization and precipitation, and gradually decreased due to subsequent Ostwald ripening and crystal growth.[38] For initial stages, the PL intensity of the PhBr-4PACz film exceeded that of the Me-4PACz film, attributed to the greater nucleation density. During the stage of crystal growth, Me-4PACz showed a rapid quenching of PL intensity due to self-aggregation, resulting in reduced crystal growth duration (68 s). On the contrary, the PhBr-4PACz film displayed the more significant PL intensity throughout all crystallization stages and had the more extended crystal growth duration up to approximately 100 s. Scanning electron microscopy (SEM) images of the buried surface (In this study, the perovskite films were exfoliated from the substrate using ultraviolet-curable adhesive to obtain an intact buried bottom surface) and cross-sections revealed that the PhBr-4PACz film exhibited significantly larger and more uniform grain sizes compared to the Me-4PACz film, confirming that its delayed crystallization kinetics and uniform micelle distribution (Supplementary Fig. 24) are beneficial to the grain growth. Additionally, no residual PbI_2 or voids were observed at the buried surface of the PhBr-4PACz film, suggesting that the higher enrichment of PhBr-4PACz at the bottom interface enhances adhesion, thereby mitigating defects at the buried surface. We further evaluated the adhesion strength of perovskite (SAM) films to the substrate through peel testing (Supplementary Fig. 25). After treatment with ultraviolet-curable adhesive, all perovskite films were successfully detached from the substrate. The Me-4PACz film was completely removed after 20 seconds of UV curing, whereas the PhBr-4PACz film required 60 seconds under the same conditions. To further quantify the interfacial adhesion force, tensile force was applied to the upper and lower glass ends of the device using a stretching device at a uniform rate (0.1 mm min^{-1}) to record the break point (Supplementary Fig. 26). The results show that the interfacial break of the Me-4PACz-based co-deposited film occurs when the tensile load increases to 385 N. In contrast, the tensile load for the PhBr-4PACz-based co-deposited film increases to 464 N, even exceeding the 433 N observed for the sequential deposition process. These experimental results confirm that the PhBr-4PACz-based co-deposition process enhances the adhesion at the perovskite/substrate interface, as expected. Consequently, the grazing-incidence wide-angle X-ray scattering (GIWAXS) measurements (Supplementary Fig. 27) revealed that the PhBr-4PACz film exhibited superior crystallization quality. In summary, the phosphate group of SAMs can accelerate the nucleation via coordination strength and coupled with large colloidal size in precursor solution triggered by the self-aggregation of SAMs, the resulting vertical distribution of SAMs are towards both interfaces in co-deposited perovskite. PhBr-4PACz due to its sterically bulky design and phenyl group can reduce the self-aggregation of SAMs in solution and thereby promote the redistribution of SAMs towards bottom interface with improved adhesion.

Hole Extraction at the Bottom Interface

So far, PhBr-4PACz has been shown to reduce self-aggregation, assemble at the bottom interface, and improve the interfacial adhesion for the co-deposited perovskite layer. Next, hole extraction at the bottom interface was investigated. In the co-deposited devices, energy loss occurs mainly at the bottom interface due to the direct contact between the perovskite and substrate without SAM coverage, which leads to a multi-fold increase in non-radiative recombination from the energy level mismatch at the bottom interface.[3,4] To monitor SAM coverage on the substrate, DMF/DMSO washing was employed to etch the co-deposited perovskite film, leaving the monolayer-anchored SAM intact, followed by XPS quantitative analysis and conductive atomic force microscopy (C-AFM) test. SEM images of the co-deposited perovskite films after washing with DMF/DMSO revealed no residual perovskite components (Supplementary Fig. 28). In the XPS spectra, the Sn *3d* peak corresponds to the substrate, while the N *1s* and P *2p* peaks correspond to the self-assembled monolayer (SAM). The calculated N/Sn area ratios for PhBr-4PACz and Me-4PACz films were 0.0068 and 0.0058, respectively, with the P/Sn ratios being 0.0014 and 0.0019, respectively. These results collectively indicate a denser accumulation of PhBr-4PACz anchored to the substrate (Fig. 2a and Supplementary Fig. 29; Supplementary Table 1). In addition, the elemental mapping from energy-dispersive X-ray spectroscopy (EDS) (Supplementary Fig. 30) indicates that the co-deposited SAMs successfully bond to the substrate, with PhBr-4PACz exhibiting a denser distribution. C-AFM was used to investigate the current distribution of the insulating SAM on the conductive substrate. As shown in Fig. 2b, the exposed conductive substrate shows a prominent dark purple high-current region, while the PhBr-4PACz substrate exhibits a less extensive dark purple area, indicating reduced substrate exposure. The current distributions for PhBr-4PACz and Me-4PACz substrates followed approximately Gaussian profiles, with median currents of 79.8 pA and 55.8 pA, respectively (Supplementary Fig. 31). This is associated with the larger dipole moment of PhBr-4PACz, which promotes interfacial carrier transport.

The PL mapping conducted through the FTO substrate reveals non-radiative recombination at the bottom interface. As depicted in Supplementary Fig. 32, the PhBr-4PACz-based co-deposited film exhibits enhanced and more uniformly distributed PL intensity compared to both the Me-4PACz-based co-deposited film and the PhBr-4PACz-based sequentially deposited film, which can be attributed to improved interfacial adhesion and SAM coverage. XPS analysis performed on the bottom surface of the perovskite film after detaching the FTO substrate reveal that the O/Pb ratio of the PhBr-4PACz-treated perovskite film is 0.0698, which is notably exceeds the corresponding ratio of 0.0346 for the Me-4PACz-treated film (Supplementary Table 2). Additionally, the O *1s* spectra shows an increase in the P=O binding energy for the PhBr-4PACz sample, with the appearance of a distinct P=O···Pb signal at 529.8 eV (Fig. 2c).[4,10] These observations suggest that vertical SAM redistribution during the crystallization process results in enhanced chemical adsorption via the phosphonic acid (PA) group at the bottom surface of the perovskite layer. Density functional theory (DFT) calculations (VASP, Fig. 2d and Supplementary Fig. 33) reveal comparable adsorption energies (~-1.1 eV) for PA groups in both SAMs, where they coordinate with Pb via P=O bonds. However, for the terminal group, compared to the methyl group (-0.22 eV), the electron-rich nature of the Br group facilitates its coordination with Pb, exhibiting a significantly stronger adsorption energy of -0.45 eV, which is further supported by an increase in the Br *3d* binding energy observed in the XPS spectra (Fig. 2d and Supplementary Figs. 34 and 35). Therefore, PhBr-4PACz, with its multiple coordination capabilities through both the Br and PA groups, significantly enhances the chemical adsorption with perovskite. Consequently, relative to the Me-4PACz-treated film, the Pb *4f* and I *3d* binding energies of the PhBr-4PACz-treated film decrease significantly by 0.30 eV and 0.25 eV, respectively (Figs. 2e, f). Notably, Me-4PACz-treated perovskite films display two weak peaks at 141.7 eV and 136.9 eV, which correspond to metallic lead (Pb^0) generated by the oxidation of iodide, a characteristic signal of surface iodide vacancy (V_I) defects.[39] The optimized geometric structure calculations demonstrate that the PA and Br groups, in proximity to the V_I site, are able to coordinate with Pb ions, forming coordination bonds with binding energies of approximately -1.28 eV and -1.20 eV, respectively. Therefore, the PhBr-4PACz with multi-site coordination effectively suppresses the formation and migration of V_I (Fig. 2g and Supplementary Fig. 36), and the bottom of perovskite likely benefits from a significant p-type doping effect due to the efficient elimination of n-type defects (V_I) by the accumulated PhBr-4PACz. Similarly, the binding energies resulting from the coordination of PA and Br with Pb_I are -1.55 eV and -1.29 eV, respectively, further highlighting that, compared to Me-4PACz, the introduction of the Br group led to the multi-site passivation of PhBr-4PACz, offering greater potential to eliminate charged defects and more effectively suppress non-radiative recombination at the GBs (Fig. 2g and Supplementary Fig. 37).[40] The trap density of states (tDOS) of the devices was measured using thermal admittance spectroscopy (Fig. 2h). The deep trap region (0.4-0.52 eV) represents defects near the bottom GBs, and the tDOS of the PhBr-4PACz-treated devices is about 5 times lower than that of Me-4PACz-treated devices, further indicating that the introduction of PhBr-4PACz significantly suppresses the deep-level defects at the GBs.

Then, UV-vis absorption and UPS spectra were carried out to examine the p-type doping level at the perovskite bottom and the energy level alignment at the perovskite/substrate interface (Supplementary Figs. 38 and 39). The Tauc plot derived from the absorption spectrum indicates a bandgap of 1.54 eV for the perovskite film (Supplementary Fig. 38b). The work function of the Me-4PACz-treated perovskite bottom was 4.88 eV, while the corresponding value for PhBr-4PACz was further increased to 5.09 eV. These trends were consistent with the changes in potential observed in the KPFM results, indicating that the PhBr-4PACz-based perovskite exhibits strong p-type characteristics. (Supplementary Fig. 40). The Me-4PACz and PhBr-4PACz-based perovskite films, cleaned using DMSO/DMF, exhibit different Fermi levels (−4.73 eV and −5.18 eV) and HOMO levels (−5.29 eV and −5.66 eV) (Supplementary Fig. 41). These trends were further corroborated by KPFM measurements (Supplementary Fig. 42) and align with previous DFT calculations. In addition, both the PhBr-4PACz-modified perovskite buried interface and the FTO substrate exhibit a more uniform surface potential compared with the Me-4PACz-modified samples (Supplementary Figs. 40 and 42). This agrees with the trends observed in XPS, C-AFM, and SEM analyses, further demonstrating that the PhBr-4PACz-based co-deposition process improves interfacial coverage and thereby results in better interfacial homogeneity. Based on these efforts, the energy level diagrams of the perovskite and SAMs were presented in Figs. 2i, j. Due to the accumulation of PhBr-4PACz at the bottom, the buried interface of the co-deposited film presents an upward-bent energy band, and the hole extraction barrier is reduced to only 0.11 eV, thereby facilitating hole extraction. Furthermore, p-type doping causes the Fermi level at the perovskite bottom to approach that of the substrate, which is expected to enhance the quasi-Fermi level splitting (QFLS). Photoluminescence quantum yield (PLQY) measurements were employed to further determine the QFLS value (Supplementary Fig. 43 and Supplementary Note 1). As depicted in Fig. 2k and Supplementary Table 3, compared to Me-4PACz, the PLQY of PhBr-4PACz-treated perovskite increases from 1.4% to 4.3%, with the corresponding QFLS value increasing from 1.17 eV to 1.20 eV, and the non-radiative open-circuit voltage loss decreasing by 29 meV. These results indicate that the vertical SAM redistribution strategy, based on PhBr-4PACz, significantly enhances the quality of the interface heterojunction, thereby leading to a considerable reduction in the open-circuit voltage loss at the interface.

When light impinges on a perovskite material, a series of physical processes occur over different timescales. Initially, excess energy leads to the formation of hot carriers, which rapidly diffuse and cool down to the band edge within a few picoseconds. Once at the band edge (cold), the photo-generated carriers continue to diffuse spatially through the perovskite material, accompanied by recombination, until they are fully extracted from the interface contact. Therefore, it is essential to evaluate the hole diffusion and extraction processes through multi-scale carrier dynamics in order to comprehensively assess the advancements achieved by the PhBr-4PACz-based vertical SAM redistribution strategy. Firstly, TRPL tests indicated that for the PhBr-4PACz-treated perovskite film, the $\tau 2$ value increased from 590.8 ns to 1600.1 ns and the $\tau 1$ value decreased from 34.6 ns to 25.1 ns, indicating lower non-radiative recombination levels and faster carrier extraction (Supplementary Fig. 44, Supplementary Table 4 and Supplementary Note 2). As TRPL lacks surface sensitivity, confocal microscopy was employed to further investigate the carrier diffusion at the interface. Fig. 2l shows the hole diffusion coefficients (D) at the bottom interface, determined by Gaussian fitting and a classical diffusion model (Supplementary Fig. 45 and Supplementary Note 3).[41] The D value of the Me-4PACz-treated film was found to be 0.07

cm^2/s, while for PhBr-4PACz, it increased to 0.13 cm^2/s, indicating that the PhBr-4PACz-treated film more effectively mitigates trapping effects and facilitates more efficient cold hole extraction. Through transient spectroscopy techniques, a deeper understanding of hole extraction was achieved. In the transient absorption spectroscopy (TAS) tests, both the 2D pseudo-color plot and normalized TAS spectra (Figs. 2m, n, and Supplementary Fig. 46) reveal the thermal relaxation of hot carriers. The high-energy spectral broadening, visible as a bright-red region in the pseudo-color plot and as a pronounced high-energy tail in the normalized spectra, directly captures the non-equilibrium distribution of hot carriers generated above the bandgap. As the delay time increases, this high-energy feature narrows and redshifts, reflecting the cooling of hot carriers as they relax toward the band edge through carrier-phonon scattering. By fitting the high-energy region with a Maxwell-Boltzmann distribution, the variation in hot carrier temperature (Tc) over time was obtained, and the hot carrier cooling time was quantified (Supplementary Fig 47 and Supplementary Note 4). The results showed that the hot carrier cooling time for the Me-4PACz-based sample was 241 fs. For PhBr-4PACz-based sample, hot carrier cooling time was decreased to 223 fs, indicating faster hot hole extraction, which can be attributed to the construction of a high-quality interface heterojunction.[17,42] In contrast, when the perovskite was peeled off from the FTO substrate, the hot carrier cooling time further increased to 395 fs, which can be attributed to the p-type doping effect and reduced defect energy levels (Supplementary Figs. 48 and 49).[43-46] Further calculations revealed that the hot carrier diffusion length increased from the Me-4PACz sample's 5 nm to 7 nm for the PhBr-4PACz sample, enhancing the likelihood of extracting hot holes at the interface (Supplementary Note 5).[47] Transient reflectance spectroscopy (TRS) was further employed to accurately evaluate hole extraction at the interface on a short time scale (Figs. 2o, p). By controlling a detection depth of approximately 20 nm and globally fitting the carrier dynamics derived from pump pulses at three different wavelengths, we determined the carrier diffusivity (D) and the buried interface extraction velocity (SEV) (Supplementary Note 6).[48,49] In comparison to Me-4PACz-treated sample, the SEV for PhBr-4PACz increased from 1569±204 cm/s to 3999±132 cm/s, indicating a significantly accelerated hole transfer process, which may also be associated with the enhanced hot carrier transport observed in perovskite. Moreover, the D value for PhBr-4PACz was further increased to 2.05±0.03 cm^2/s, suggesting an improvement in surface hole extraction. Apart from the effective extraction promoted by improved energy band alignment, this may also be attributed to the increased concentration of hot holes and cold holes caused by the p-type doping effect. In conclusion, multi-scale analysis of carrier dynamics demonstrates that the PhBr-4PACz-based vertical SAM redistribution strategy significantly enhances the extraction of both hot and cold holes.

In-situ crosslinking of AVIMCl to suppress SAM diffusion

While the hole extraction at the buried interface is significantly improved via introducing a SAM, e.g. PhBr-4PACz, the slight dislocation of the SAMs in the perovskite phase in the co-deposited devices can still lead to non-negligible diffusion along GBs during the aging process. Particularly, thermal stress is a key factor triggering the desorption and diffusion of SAMs.[25,27] Once SAMs diffuse and reach the top interface, the energy level matching at the top interface will be disrupted, resulting in compromised PCE. Therefore, it is imperative to address the GBs through filling or crosslinking.

According to previous studies, certain passivating molecules can diffuse downwards along the GBs from top surface to the bottom of the perovskite layer.[50-53] This post-treatment process effectively prevents interference with the vertical distribution optimization of SAM during the co-deposition process and, through in-situ crosslinking of the GBs, offers the potential to suppress SAM diffusion. Therefore, ionic liquids with superior diffusivity present a promising option for consideration. Here, AVIMCl was carefully selected and introduced to the perovskite surface. As illustrated in Supplementary Figs. 50 and 51, the glass transition temperature (T_g) of AVIMCl is -50°C. At room temperature, AVIMCl exists as a viscous liquid. However, when heated above 60°C, it exhibits favorable fluidity, which facilitates the downward diffusion of AVIMCl along the GBs. In addition, using the C=C bond as the active site for polymerization, the corresponding polymer (PAVIM) was successfully prepared by heating the AVIMCl solution to 100 °C in the presence of initiators, which was confirmed by FTIR analysis. As shown in Supplementary Fig. 52, the disappearance of the characteristic peak at 1650 cm^{-1}, attributed to the C=C stretching vibration in the polymerized product, demonstrated the successful conversion of AVIM to PAVIM. AVIMCl was introduced onto the surface of the perovskite, and FTIR analysis was performed on both the top and buried surfaces of the perovskite film to further confirm the formation of PAVIM within the perovskite film after annealing. To facilitate the detection of infrared signals from AVIMCl on the buried surface, the concentration of AVIMCl had to be increased to 20 mg/mL. As shown in the Fig. 3a, characteristic peaks observed at 1583 cm^{-1} and 1561 cm^{-1} are attributed to the imidazole backbone and can also be detected on the buried surface of the perovskite film, confirming that AVIMCl can diffuse along the GBs to the buried interface. Additionally, the C═C characteristic peak, which was clearly observed after the introduction of AVIMCl into the perovskite film, significantly weakened on both the top and bottom surfaces of the film after heating, demonstrating the in-situ crosslinking behavior of AVIMCl following its diffusion along the GBs. SEM was conducted to examine the influence of the optimal concentration of AVIMCl on the morphology of perovskite films. As shown in Fig. 3b, the white, needle-like PbI_2 crystals present on the perovskite surface, which were distributed along the GBs in the PhBr-4PACz-based perovskite film, disappeared upon the addition of AVIMCl. Additionally, the GBs of the PhBr-4PACz-based perovskite appeared to become indistinct following AVIMCl treatment. XRD analysis (Fig. 3c) demonstrated that there were no significant shifts or disappearances of the characteristic perovskite peaks. This indicates that AVIMCl did not incorporate into the perovskite lattice during diffusion along the GBs, but instead accumulated at the GBs. Moreover, the PbI_2 peak at 12.7° disappeared, while the intensity of all perovskite characteristic peaks increased. A detailed analysis of the main (100) peak further revealed a notable reduction in the full width at half-maximum (FWHM) from 0.175° to 0.111° after AVIMCl treatment (Supplementary Fig. 53), indicating improved crystallinity. Therefore, the introduction of AVIMCl not only preserves the integrity of the perovskite but also exhibits passivating properties (as discussed later). ToF-SIMS analysis was employed to more intuitively examine the distribution of AVIMCl and its suppressive effect on SAM diffusion (Supplementary Fig. 54). Following the introduction of AVIMCl and in-situ crosslinking, the AVIMCl signals were primarily localized near the top surface and bottom interface, indicating that a portion of the AVIMCl introduced at the surface is capable of diffusing along the GBs to the bottom. It is noteworthy that the AVIMCl distribution at the bottom was closer to the bulk phase than that of PO_2^-, indicating that the AVIMCl at the bottom primarily accumulates near the bottom GBs rather than at the bottom surface. To further confirm the distribution of AVIMCl, XPS and KPFM tests, which are sensitive to surface information, were employed (Supplementary Figs. 55 and 56). In the XPS N *1s* spectra, after the modification with AVIMCl on the top surface of perovskite, a strong -N^+H characteristic peak corresponding to AVIMCl appeared at a binding energy of 401.9 eV. However, no -N^+H peak was observed on the bottom surface of perovskite, indicating that AVIMCl is almost absent on the buried surface of perovskite. Furthermore, the surface potential of the bottom interface of perovskite only increased from 69.0 mV to 74.6 mV after AVIMCl modification, showing a change that is negligible. This indicates that the introduction of AVIMCl does not significantly affect the p-type doping level of PhBr-4PACz at the bottom surface, which could also be related to the fact that PhBr-4PACz at the GBs blocks the further downward diffusion of AVIMCl. Most importantly, after thermal stress aging, the PhBr-4PACz film exhibited a significant PO_2^- signal in both the bulk phase and top regions, while the PO_2^- signal near the bottom was noticeably reduced (Figs. 3d-e). These findings suggest that a significant amount of SAMs diffused in the bulk phase from the bottom to the top interface. In contrast, the aged film treated with AVIMCl showed no obvious PO_2^- signals in the bulk phase, and the PO_2^- signal from the bottom side remained nearly constant, confirming that the exceptional effectiveness of the AVIMCl-based in-situ GB crosslinking strategy in suppressing SAM diffusion. The mechanism of AVIMCl penetrating along the GBs and undergoing GB crosslinking to suppress SAM diffusion from the bottom to the top surface of perovskite is illustrated in Fig. 3f.

The passivation characteristics and stress regulation of AVIMCl

Although the introduction of AVIMCl successfully suppressed the upward diffusion of SAMs along the GBs under thermal stress aging, the passivation characteristics of AVIMCl and the crosslinking strategy's impact on stress regulation in the perovskite layer remain to be clarified. As shown in Figs. 4a, b, after AVIMCl treatment, the perovskite films transitioned from a red-white mixed appearance to a uniform blue, corresponding to an increase in PL intensity and an improvement in the average carrier lifetime (from 556 ns to 1530 ns), indicating a significant reduction in the non-radiative recombination on the perovskite surface. The interaction mechanism between AVIMCl and the perovskite was further analyzed using XPS. As depicted in the Supplementary Fig. 57, after AVIMCl modification, the characteristic peak of Pb *4f* shifted to a lower binding energy region, which can be attributed to the nitrogen atom on the imidazole providing a lone pair of electrons to form a coordination bond with the uncoordinated Pb^{2+}. Additionally, electrostatic potentials tests reveal that imidazole cations exhibit electron-withdrawing properties (Supplementary Fig. 58). The accumulation of AVIMCl on the perovskite surface is expected to create an electron-deficient surface, thereby reducing the work function of the perovskite and promoting electron extraction. KPFM images show that after the introduction of AVIMCl, the surface potential of the perovskite increased from 235 mV to 451 mV, confirming the upward shift of the Fermi level (Fig. 4c). The band structure obtained from UPS further confirms that AVIMCl induced a more n-type surface, which facilitates electron transfer and extraction to PCBM (Fig. 4d and Supplementary Fig. 59). To further quantify the reduction in non-radiative recombination losses due to the AVIMCl strategy, the QFLS of the corresponding samples was calculated using PLQY data (Supplementary Table 3). After AVIMCl modification, the interface non-radiative recombination loss was reduced by 6 mV, and the QFLS increased from 1.20 eV to 1.21 eV, which helps further improve V_{OC}.

To evaluate the impact of the AVIMCl-based crosslinking strategy on the mechanical properties of the perovskite, Peak Force Quantitative Nanomechanical Atomic Force Microscopy (PFQNM-AFM) was first used to assess the Young's modulus of various samples. As shown in the

Fig. 4e, after the introduction of AVIMCl, the average Young's modulus of the perovskite films decreased from 7.66 GPa to 4.75 GPa, contributing to a more uniform stress distribution and reduced local strain. Therefore, grazing-incidence X-ray diffraction (GIXRD) technique with the 2θ-$\sin^2\varphi$ method (Supplementary Note 7) was used to further analyze the effect of GB crosslinking on residual strain.[54,55] By fixing the 2θ angle and adjusting the incident angle (ψ), two characteristic detection depths of the perovskite films were selected, corresponding to incident angles of 0.2° and 0.5°, which correspond to detection depths of 200 nm and 500 nm, respectively. The results (Fig. 4f and Supplementary Fig. 60) showed that the PhBr-4PACz-based perovskite films exhibited residual tensile stress of 19.45 MPa and 10.43 MPa at the 200 nm and 500 nm depths, respectively. After introducing AVIMCl, the scattering peaks shifted from a significant rightward shift to a slight leftward shift, and the residual stress transitioned to compressive stress of -4.43 MPa and -2.23 MPa, respectively. This suggests that AVIMCl, through crosslinking on both surfaces, more effectively released the internal tensile stress, helping to reduce defects and traps at the crystal center and enhancing the long-term stability of the device. [56-57]

Device photoelectric performance and stability

To emphasize the advantages of the vertical SAM redistribution strategy based on PhBr-4PACz and the in-situ crosslinking strategy based on AVIMCl, the photovoltaic performance and stability of devices (FTO/perovskite (PhBr-4PACz)/AVIMCl/PCBM/BCP/Cu) were tested. Based on performance testing of PSCs with varying concentrations of added molecules (Supplementary Figs. 61 and 62; Supplementary Table 5), we identified 2.0 mM for PhBr-4PACz as the optimal concentrations for the fabrication of best-performance devices, which is higher than Me-4PACz (1.5 mM). The reason for this phenomenon may be the bulky PhBr-4PACz, which provides more comprehensive passivation sites and reduces the damage caused by excessive SAM aggregation to the perovskite crystal quality. The current density–voltage (*J-V)* curves of the champion devices (Fig. 5a and Supplementary Table 6) show that the PhBr-4PACz-modified film effectively enhances all parameters. The enhancement in current density (J_{SC}) and fill factor (FF) can be attributed to the improved adhesion at the bottom and the p-type doping effect induced by the bromine group and bulky steric hindrance of PhBr-4PACz. The increase in V_{OC} is due to the reduced conduction band offset and enlarged QFLS. Moreover, the introduction of AVIMCl further enhances V_{OC} and FF, owing to the formation of an n-type upper surface, the suppression of non-radiative recombination at the upper surface, and the reduction of residual stress in the bulk phase (Fig. 5a; Supplementary Fig. 63; Supplementary Tables 6 and 7). Ultimately, the device with PhBr-4PACz/AVIMCl elevates PCE from 23.91% to 27.26%, with an V_{OC} of 1.193 V, J_{SC} of 26.28 mA cm^{-2}, and FF of 86.95%. The stable power output (SPO) at the bias voltage of the maximum power point under AM 1.5G for 300 s were 26.78%, 25.57%, and 24.37% for PhBr-4PACz/AVIMCl, PhBr-4PACz and Me-4PACz devices, respectively. (Fig. 5b). The PCEs of 30 devices are presented as box plots in Fig. 5c. The PhBr-4PACz/AVIMCl-modified, PhBr-4PACz-modified, and Me-4PACz-modified devices achieved average efficiencies of 26.86% ± 0.16%, 25.73% ± 0.22%, and 23.31% ± 0.32%, respectively, indicating that the PhBr-4PACz/AVIMCl-modified device exhibits the most concentrated efficiency distribution and superior reproducibility.[58-59] Additionally, the external quantum efficiency (EQE) results (Fig. 5d) show that the integrated J_{SC} for the PhBr-4PACz/AVIMCl-modified device is 25.15 mA/cm², with an error within 5% with J_{SC} of *J-V* curve, meeting industry standards. Notably, one PhBr-4PACz/AVIMCl-based device was certified by a third-party independent institution (NPVM, National PV industry measurement and testing center) to achieve a PCE of 27.03% in Supplementary Fig. 64. After 300 seconds of maximum power point (MPP) tracking, the quasi-steady-state PCE reached 26.50% (Supplementary Fig. 64), certified by NPVM. We also compared our strategy with other reported IPSCs using a co-deposition or sequential deposition process with a bandgap approaching 1.55 eV for perovskite absorber layer, and PhBr-4PACz/AVIMCl-based device represents one of the highest efficiencies among these IPSCs (Fig. 5e; Supplementary Tables 8 and 9). Furthermore, we fabricated IPSCs on other TCO substrates to test the versatility of our strategy. We fabricated IPSCs on relatively smooth ITO substrates, modified with Me-4PACz, PhBr-4PACz, and PhBr-4PACz/AVIMCl in Supplementary Fig. 65 and Supplementary Table 10. The highest PCE achieved for the PhBr-4PACz/AVIMCl-based device was 26.55%, slightly lower than the PCE on FTO substrates. The primary difference in J_{SC} can be attributed to the textured structure of FTO, which reduces light scattering losses compared to ITO. Additionally, we also fabricated flexible perovskite solar cells (FPSCs) with the following configuration: PET/ITO/perovskite (PhBr-4PACz)/AVIMCl/C60/BCP/Cu. The 0.075 cm² FPSC achieved a PCE of 25.03% (The certified efficiency under reverse scan and the steady-state output were 24.49% and 24.41%, respectively). (Fig. 5f and Supplementary Fig. 66). We also fabricated the corresponding perovskite films using co-deposition or sequential deposition techniques on FTO substrates with an area of 5×5 cm² (Supplementary Fig 67). The perovskite films prepared by the co-deposition technique exhibited better substrate coverage, especially for Me-4PACz, which has poorer wettability when used alone. In Supplementary Fig 68, the mini-module based on the dual modification strategy of PhBr-4PACz/AVIMCl (with an aperture area of 8.905 cm²) achieved a PCE of 23.68%, with a V_{OC} of 5.841 V, a J_{SC} of 4.91 mA cm^{-2}, and an FF of 82.54%. These results demonstrate the high commercial application value of our strategy, as it can adapt to various substrates and is suitable for large-area applications.

Stability is one of the most crucial indicators for determining the commercial potential of PSCs. We investigated the thermal stability of unencapsulated devices by aging them at 85°C in nitrogen (Fig. 5g). The devices modified with PhBr-4PACz/AVIMCl maintained over 90% of their initial PCE after 2000 hours of thermal aging, outperforming those modified with PhBr-4PACz (72.3%). Devices modified with Me-4PACz dropped to 61.9% of their original PCE after just 1032 hours of aging. The unencapsulated devices underwent maximum power point tracking (MPPT) under 1-sun illumination in an N_2 glovebox at 65°C (Fig. 5h) according to the provided ISOS-L-2I standard protocol. After 2000 hours of MPPT, the PhBr-4PACz/AVIMCl-based PSC retained 96.6% of its initial PCE, while those based on PhBr-4PACz and Me-4PACz dropped to 85.2% and 77.3%, respectively. Additionally, we studied the mechanical stability of flexible devices under repeated bending cycles (Supplementary Fig. 69). After 5000 bending cycles (r= 5mm), the Me-4PACz-based, PhBr-4PACz-based, and PhBr-4PACz/AVIMCl-based devices maintained 59.8%, 85.1%, and 94.7% of their initial PCE, respectively. The optimized vertical SAM distribution and the more comprehensive passivation characteristics of PhBr-4PACz, along with improved interfacial adhesion, significantly enhanced the stability of both rigid and flexible devices compared to Me-4PACz. Simultaneously, in-situ crosslinking suppresses the diffusion of the bottom SAM, reduces residual stresses in the bulk phase, and minimizes defects on the top surface. This further improves the performance and stability of these devices, providing insights into the commercialization of perovskite photovoltaics.

Discussion

In our work, we have proposed a method to manage the vertical distribution of co-deposited SAM, including SAM design and GB cross-linking strategies, to break the bottleneck in the efficiency of co-deposited PSCs. Specifically, PhBr-4PACz was added to the precursor solution, which suppress the face-to-face stacking of the conjugated core through the large dihedral angle, alleviating SAM self-aggregation, and reduces damage to the perovskite crystallization and enhances its adhesion at the bottom interface. Moreover, the introduction of Br provides more comprehensive passivation sites and induces a p-type doping effect at the bottom of the perovskite layer. As a result, a high-quality heterojunction is formed at the bottom of the perovskite layer, improving the extraction of both hot and cold holes. Subsequently, AVIMCl was introduced to the top of the perovskite to perform in-situ cross-linking, which penetrates along the GBs to suppress the diffusion of SAM at the bottom interface, reduce residual stress in the bulk phase, and minimize recombination losses at the interface. Ultimately, the p-i-n structured PSCs based on PhBr-4PACz/ AVIMCl achieved a champion PCE of 27.26% (certified at 27.03%), which is the highest reported PCE for co-deposited devices to date. In stability tests, the PSCs based on PhBr-4PACz/AVIMCl retained more than 90% of their initial efficiency after 2000 hours of aging at 85°C in a dark environment. Under ISOS-L-2 standard protocol, the efficiency remained at 96.6% after 2000 hours of maximum power point tracking (MPPT) at 65°C. Furthermore, we achieved a maximum PCE of 26.55% on ITO substrates, as well as 25.03% (certified at 24.49%) for small-area flexible PSCs on PET/ITO substrates, demonstrating the versatility of these strategies across different substrates. Our approach provides insights for advancing the development of high-performance perovskite solar cells.

Methods

Materials

All chemicals were used as received without further purification. Formamidine iodide (FAI, >99.5%) was gained from Greatcell Solar Materials Pty Ltd. Cesium iodide (CsI, >99.99%), formamidine chloride (FACl, >99.5%), formamidine chloride, methylammonium chlorine (MACl, >99.5%), Bathocuproine (BCP), (4-(3,6-dimethyl-9H-carbazol-9-yl)butyl)phosphonic acid (Me-4PACz) and (4-(3-bromo-6-phenyl-9H-carbazol-9-yl)butyl)phosphonic acid (PhBr-4PACz) were purchased from Xi'an Yuri Solar Co., Ltd. [6,6]-Phenyl-C61-butyric acid methyl ester (PCBM) was gained from Lumtec, Taiwan. Lead (II) iodide (PbI_2, >99.99%) was gained from Xi'an E-Light New Material Company Limited. N, N dimethylformamide (DMF, 99.8%), dimethyl sulfoxide (DMSO, 99.8%), isopropanol (IPA, 99.5%), chlorobenzene (CB), 3,6-

dimethyl-9H-carbazole (Me-Cz), 3-bromo-6-phenyl-9H-carbazole (PhBr-Cz) and methanol (MeOH, 99.9%) were obtained from Sigma Aldrich Chemical Company Limited. 1-Allyl-3-vinylimidazolium chloride (AVIMCl) was gained from McLin Biochemical Technology Co., Ltd. 2,2'-Azobis(2-methylpropionitrile) (AIBN) and ammonium persulfate (APS) were obtained from Grate (Shanghai) Technology Co., Ltd. The FTO substrate (model number: HM FTO-22), purchased from Beijing Huamin New Materials Technology Co., Ltd, possesses a square resistance of 8 ohms, and a thickness of 2.2 mm. An anti-reflective film (BA4076, Japan) was also placed on the back of the FTO glass.

Co-deposited perovskite film fabrication

First, FAI, PbI_2, CsI, MACl and FACl were fully dissolved in mixed solvents of DMF and DMSO (4:1, v/v) according to the stoichiometric formula of 1.5M $Cs_{0.05}(FA_{0.90}MA_{0.05})PbI_3$ in anhydrous DMF: DMSO =4:1 (v:v). Then, the solution was stirred at room temperature until dissolved. PhBr-4PACz was then added to the perovskite precursor at a concentration of 2mM, followed by shaking for 300 s. The precursor solution was filtered through a 0.22 μm polytetrafluoroethylene filter before use. 60 μL perovskite precursor solution was dropped onto substrates and spin-coated at 1000 rpm for 25 s and 5000 rpm for 30 s. The 120 μL of anisole is poured on the substrates 7 s prior to the end of the program and then the substrates are annealed at 100 °C for 15 min.

Fabrication of integrated devices

The glass/FTO substrate underwent a cleaning process, which included ultrasonic treatment in sequential baths of anhydrous ethyl alcohol, detergent, deionized water, acetone, and isopropanol. Afterward, the substrate was dried and then subjected to a 15-minute exposure in a UV-ozone environment to remove organic contaminants. Then, FTO substrates were transferred into the nitrogen-filled glovebox quickly maintaining less than 0.01 ppm of O_2 and H_2O. After cooling down to room temperature, perovskite with SAM solution was deposited on the substrate as detailed above. Subsequently, a 0.5 mg/mL AVIMCl IPA solution with 0.1mol% AIBN and APS was spin-coated on the perovskite layer at 5000 rpm for 30 seconds and annealed at 100 °C for 30 minutes. Then, PCBM (20 mg/mL in CB) was spin-coated at 2000 rpm for 30 s and BCP (saturated solution in IPA) was spin-coated at 5000 rpm for 30 s. Finally, Cu (120 nm) is deposited as back electrode through a mask by thermal evaporation under 9.9×10^{-5} Pa. The device area is confined to 0.0737 cm^2 by a mask.

Fabrication of mini-modules

The FTO substrates (5 cm × 5 cm) were ablated P1 patterns by a laser scriber and then cleaned with same process of small-area device. Then, FTO substrates were transferred into the nitrogen-filled glovebox quickly. After cooling down to room temperature, 1 mL perovskite with SAM solution was deposited on the substrate as detailed above. The 400 μL of chlorobenzene was poured on the substrates 10 s prior to the end of the program (Note: For 1-mL pipette tips, the distal 1 cm should be removed during use). Then, quickly transferred the substrates to annealing at 100 °C for 10 min. Then, 200 μL AVIMCl solution was spin-coated on was deposited on the substrate as detailed above, and annealed at 100 ℃ for 30 min. Then, the C60 (27 nm) and BCP (7 nm) were sequentially deposited on the perovskite films by vacuum evaporation. The P2 lines were applied adjacent to the P1 lines using laser scribing. Finally, Cu (120 nm) was deposited as back electrode by thermal evaporation under 9×10^{-5} Pa and then P3 lines were scribed. The aperture area of the module is 8.905 cm^2.

Exfoliation of the perovskite films

A simple delamination structure was made using UV-curing glue and a cover glass, and then the perovskite film was completely delaminated under sufficient curing time, as shown in Supplementary Fig. 25.

Characterization

The NMR spectra are recorded on Bruker Avance III (600 MHz). X ray photoelectron spectroscopy (XPS) was acquired by an AXIS Ultra DLD X-ray photoelectron spectrometer and calibrated against the C *1s* peak (284.8 eV). Grazing incidence wide-angle X-ray scattering (GIWAXS) tests are performed at beamline BL14B in the Shanghai Synchrotron Radiation Facility (SSRF) with an X-ray wavelength of 0.124 nm, the GIWAXS patterns are acquired with a grazing incidence angle of 0.3º and an exposure time of 20 s. XRD was tested by Mini Flex 600 X-ray diffractometer (with Cu-Ka radiation as X-ray source) with a scan rate of 15° min^{-1}. Time-of-flight secondary ion mass spectra (ToF-SIMS) were detected by ION TOF ToF SIMS 5-100 (Primary ion beam: Bi^{3+}, 30 keV, incident angle: 45 deg, scanning area: 150*150 um^2, pixel: 128*128, beam current: 0.48 pA). Grazing-incidence X-ray diffraction (GIXRD) was performed using a SmartLab X-ray diffractometer. The data for the single crystals of Me-Cz and PhBr-Cz were collected with a D8 VENTURE Single crystals diffractometer with a large area photon II detector. The single crystals used for X-ray diffraction analysis were obtained by slow evaporation in THF. Ultraviolet and visible (UV-vis) spectrophotometry was tested by EV 300 (Thermo Fisher Scientific). PL mapping images，PL mapping images and cold carrier diffusion data were obtained by MicroTime 200. Steady-state photoluminescence (PL) and time-resolved photoluminescence (TRPL) were performed using an FLS 1000 photoluminescence spectrometer. Photoluminescence quantum yield (PLQY) was measured using an LuQY Pro LP20-32 system under 1-sun equivalent excitation. The samples were excited inside an integrating sphere by a 532-nm laser, whose intensity was calibrated to 1-sun conditions using a perovskite solar cell, adjusting the laser so that the short-circuit current density matched the J_{SC} obtained under a standard solar simulator. The measured PLQY was then used to calculate the quasi-Fermi level splitting (QFLS). SEM images were acquired by field emission scanning electron microscope (JSM-7800F). Kelvin probe force microscopy (KPFM), atomic force microscopy (AFM) and conductive atomic force microscopy (c-AFM) images were observed on the Bruke Bio Fast Scan AFM using taping mode. Ultraviolet photoelectron spectroscopy (UPS) was measured by AXIS Ultra DLD machine under excitation from the He I line (21.22 eV) of a helium discharge lamp. The current density voltage (*J*-*V*) curves of the device were obtained under AM1.5G illumination at 100 mW cm^{-2} (calibrated by standard Si cell with quartz filter (91150V, Newport)) using an Abet Technologies Sun 2000 solar simulator and a Keithley 2400 source meter. The 91150V Reference Cell is a 2 x 2 cm^2 calibrated solar cell made of monocrystalline silicon and a fused silica window. The certification of reference cell is accredited by NIST to the ISO-17025 standard and is traceable to the National Renewable Energy Laboratory (NREL). The *J-V* curves of devices were measured in reverse (from 1.25 V to -0.1 V) and forward (from -0.1 V to 1.25 V) scan modes with a scan step length of 0.03 V and a dwell time of 10 ms for each voltage. The solar cells were tested at room temperature in a N_2-filled glove box. All statistical results were obtained from the average of 30 cells. The MPPT operational stability was evaluated under continuous illumination (AM1.5G) at the maximum power point (65°C) according to the ISOS-L-2 standard protocol. External quantum efficiency (EQE) test was performed by the QTEST HIFINITY 5 EQE system (the light intensity was calibrated with standard Si detectors). The transient absorption measurements were performed using a fiber laser (1030 nm, 100kHz repetition rate, YF-FL-10-100-IR, Yacto-Technology, China) as the laser source and a femto-TA100 spectrometer (Time-Tech Spectra, China). The pump beam was generated using an optical parametric amplifier (YactoOPA).

Calculation

Geometries optimization and frequency analysis was performed at the B3LYP2/6-311G(d,p), and a more accurate def2TZVP basis set was used for dipole moment calculations. The molecular orbitals and dipole moments were analyzed, and the energy gaps between HOMO and LUMO were obtained. These calculations were based on Density Functional Theory (DFT) with Gaussian 09 package. The calculations about the dimer and trimer system of SAMs were performed using Gaussian at the level of M06-2X/def2-TZVP with DFT-D3 method. The spin-polarized DFT calculations were performed using the Vienna Ab Initio Simulation Package (VASP) code, employing periodic boundary conditions. The projector augmented wave (PAW) pseudopotential was used to describe the ion-core electron interaction, while the exchange-correlation energy of valence electrons was described by the Perdew-Burke-Ernzerhof (PBE) functional with a plane-wave cutoff energy of 400 eV. Grimme's D3 corrections were also used to describe the dispersion interaction. The energy convergence criteria of structure optimization (including lattice parameters and internal atomic positions) were set to 10^{-5} eV for each self-consistent field (SCF) iteration step, and the maximum force convergence criterion for geometry optimization was set to 0.05 eV/Å. For surface calculations involving $FAPbI_3$, a surface/vacuum structure with a vacuum thickness of 15 Å was considered. The molecular graphics viewer VESTA was used to plot the crystal structures.

Data availability

Data supporting this study are available within the Article and Supplementary Information. Source Data Files are provided with this paper. Additional data are available from the corresponding author on request. The X-ray crystallographic coordinates for the structures reported in this study have been deposited at the CCDC under deposition numbers 2512507 and 2512509. These data can be obtained free of charge from the CCDC via www.ccdc.cam.ac.uk/data_request/cif.

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Author Contributions Statement

C.-C.C. supervised the project. R.Z. and C.-C.C. conceived the idea. R.Z., A.S., C.T., and C.-C.C. designed the research. R.Z., C.C., S.L. and X.W. carried out the fabrication of PSCs. S.H. and L.C. performed the TAS characterization. R.Z. and J.C. carried out the DFT calculations. R.Z., Q.C., Y.Z., R.L., T.X., T.C., and K.Z. contributed to provide helpful discussions and performed other experiments. R.Z., A.S. and C.T. analyzed the data. R.Z. and C.-C.C. wrote the manuscript. All authors discussed the results and commented on the manuscript.

Competing Interests Statement

The authors declare no competing interests.

Figure Legends

Fig. 1 Properties, Distribution, and Crystallization Regulation of PhBr-4PACz. **a** ESP and dipole moment of Me-4PACz and PhBr-4PACz. **b** HOMO energy levels of Me-4PACz and PhBr-4PACz. **c, d** Molecular packing patterns in single crystals corresponding to the conjugated cores of Me-4PACz (3.6-dimethyl-9*H*-carbazole) and PhBr-4PACz (3-bromo-9-methyl-6-phenyl-9*H*-carbazole). **e** DFT calculation of steric effect of the corresponding dimers of Me-4PACz and PhBr-4PACz. **f** ToF-SIMS profiles of FTO/perovskite (Me-4PACz) and FTO/perovskite (PhBr-4PACz) samples. **g** In-situ photoluminescence (PL) spectra of the perovskite (Me-4PACz) and perovskite (PhBr-4PACz) films during annealing.

Fig. 2 Hole Extraction at the Bottom Interface. **a, b** N *1s* and Sn *3d* XPS spectra and C-AFM with a scale bar of 500 nm for the FTO/SAM substrates after washing off the perovskite (SAM) films. **c** XPS spectra of O *1s* for buried surface of perovskite (SAM) films. **d** The adsorption energies of the anchoring groups and terminal groups of various SAMs with perovskite surface. **e, f** XPS spectra of Pb *4f* and I *3d* for the buried surface of perovskite (SAM) films. **g** The binding energies of the anchoring groups and terminal groups of SAMs with Iodine vacancies (V_I) and lead–iodine antisites (Pb_I) defects in perovskite. **h** tDOS curves of the PSCs. **i, j** Energy level diagrams of FTO/perovskite (Me-4PACz) and FTO/perovskite (PhBr-4PACz). **k** QFLS and $\Delta V_{non\text{-}rad}$ of perovskite (SAMs) films. **l** Evolution of the σ_x profile broadening as a function of time extracted from the bottom of FTO/perovskite (SAM) samples. **m, n** Normalized fs-TA spectra of perovskite (Me-4PACz) and perovskite (PhBr-4PACz) films deposited on FTO. **o, p** TRS measurements of perovskite (Me-4PACz) and perovskite (PhBr-4PACz) films deposited on FTO. TRS measurements were performed with excitation from the glass side.

Fig. 3 In-situ crosslinking of AVIMCl to suppress SAM diffusion. **a** FTIR of perovskite/AVIMCl film before and after cross-linking. **b, c** Top view SEM images with a scale bar of 400 nm, and XRD patterns of perovskite (PhBr-4PACz) films with and without AVIMCl treatment. **d, e** The ToF-SIMS profiles of FTO/perovskite (PhBr-4PACz) and FTO/perovskite (PhBr-4PACz)/AVIMCl samples after 150 hours of day-night alternating thermal aging at 85°C along with the corresponding 3D profile analysis of SAM (PO_2^-). **f** A schematic diagram illustrating in-situ crosslinking of AVIMCl to suppress SAM diffusion.

Fig. 4 The passivation characteristics and stress regulation of AVIMCl. a, b PL mapping (scale bar: 20 μm) and TRPL mapping (scale bar: 20 μm) of perovskite (PhBr-4PACz) and perovskite (PhBr-4PACz)/AVIMCl films. The images were collected from the top surface side. **c** KPFM surface potential images (scale bar: 1 μm) of perovskite (PhBr-4PACz) and perovskite (PhBr-4PACz)/AVIMCl films. **d** Energy-level diagrams of perovskite/PCBM stacks with PhBr-4PACz modified and PhBr-4PACz/AVIMCl modified perovskite films. **e** Young's moduli (scale bar: 1 μm) of perovskite (PhBr-4PACz) and perovskite (PhBr-4PACz)/AVIMCl films. **f** Linear fitting of the 2θ–$\sin^2\Psi$ curves of perovskite (PhBr-4PACz) and perovskite (PhBr-4PACz)/AVIMCl films, where the incident angles were 0.2° and 0.5°. The insets display schematics of residual stress of the perovskite at different depths; the left and right correspond to the residual stress at incident angles of 0.2° and 0.5°, respectively.

Fig. 5 Device photoelectric performance and stability. a *J*–*V* curves of the best-performing devices with Me-4PACz, PhBr-4PACz and PhBr-4PACz/AVIMCl. **b** Current density and PCE measured at the maximum power point (MPP) for 300 s of the best-performing SAM-modified PSCs. **c** Box plots depicting the distribution of PCE for devices based on Me-4PACz, PhBr-4PACz, and PhBr-4PACz/AVIMCl. Data were obtained from measurements of 30 independent devices. In the box plots, the horizontal line within each box represents the median, and the box represents the mean. **d** EQE spectra of the devices with Me-4PACz, PhBr-4PACz and PhBr-4PACz/AVIMCl. **e** Comparison of the V_{OC} and PCE of our PSCs with recently reported high-performance inverted PSCs. Including sequential deposition devices and co-deposition devices. **f** Best *J*–*V* curves of flexible PSCs**. g** Thermal stability of the unencapsulated devices heated at 85°C in N_2 under dark environment. **h** MPP tracking of the devices under ISOS-L-2 standard protocol.

Editorial summary:

Co-deposited perovskite solar cells suffer from self-assemble monolayer (SAM) aggregation that weakens interfacial adhesion and limits performance. Zhuang et al. design an asymmetric SAM and a crosslinking additive to curb aggregation, enabling efficient, stable devices.

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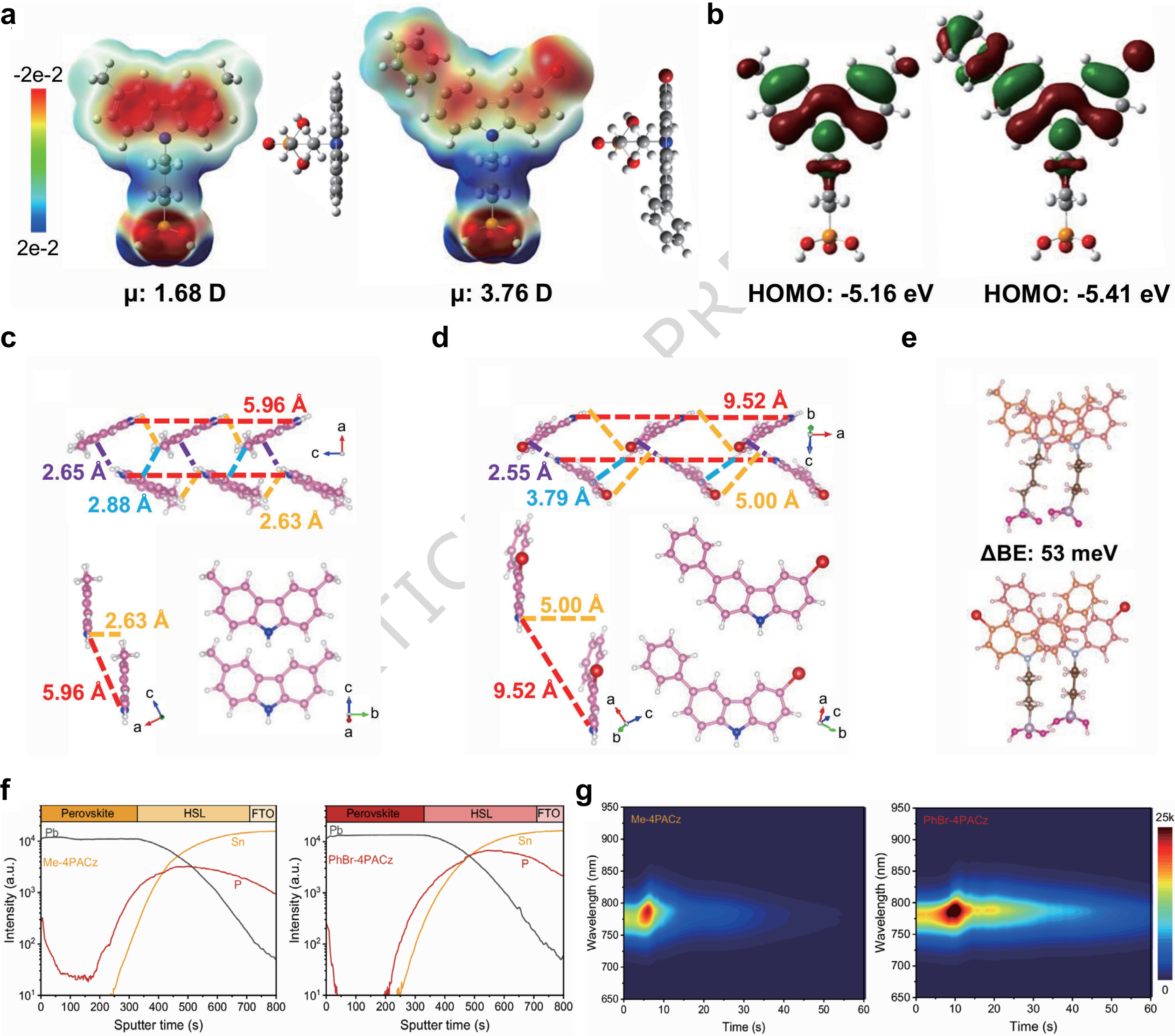
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μ: 3.76 D
b
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c
5.96 Å
2.65 Å
2.88 Å
2.63 Å
2.63 Å
5.96 Å
d
9.52 Å
2.55 Å
3.79 Å
5.00 Å
5.00 Å
9.52 Å
e
ΔBE: 53 meV
f
Perovskite
HSL
FTO
Pb
Sn
P
Me-4PACz
PhBr-4PACz
Intensity (a.u.)
Sputter time (s)
g
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PhBr-4PACz
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Time (s)
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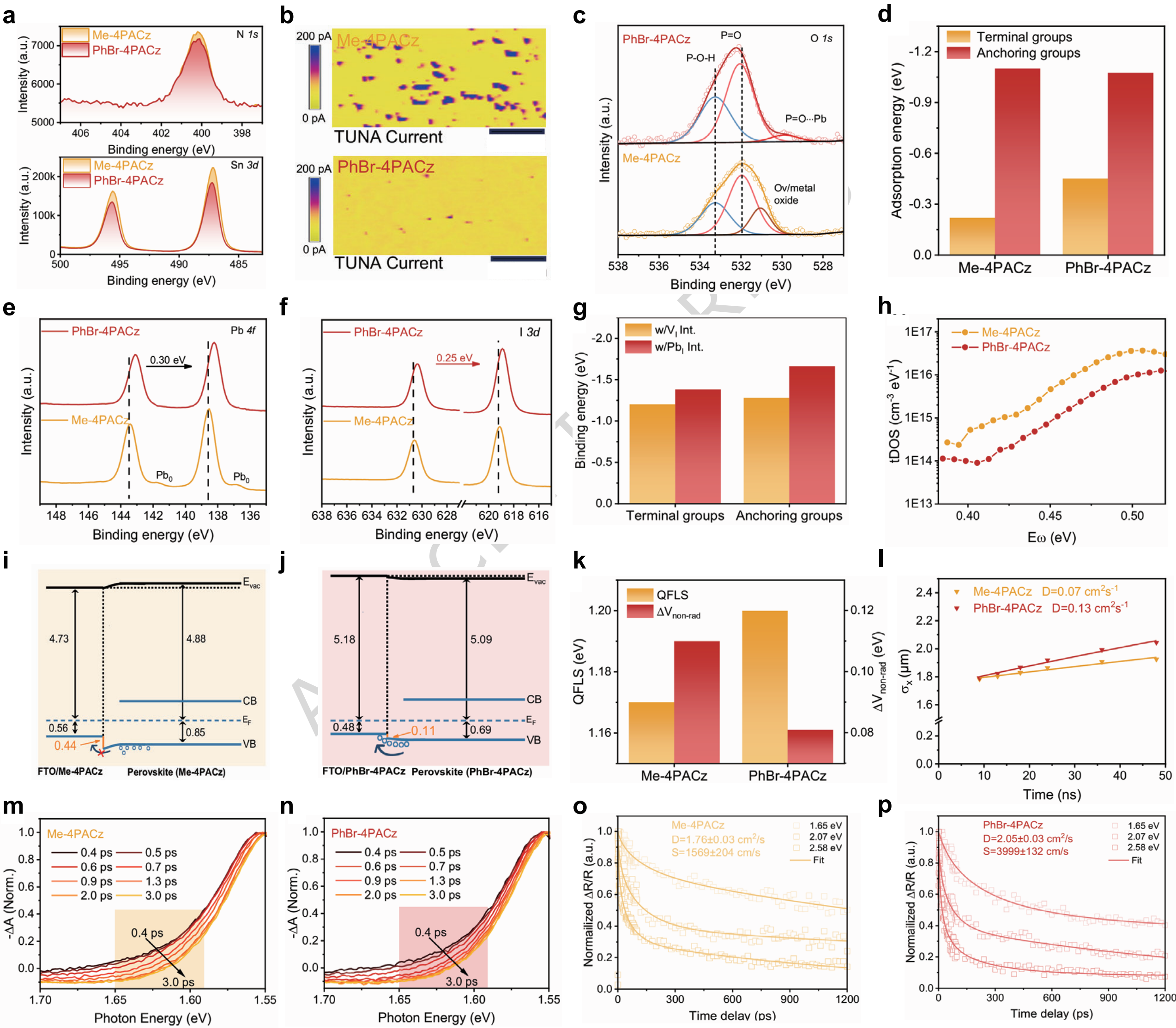
0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Me-4PACz
PhBr-4PACz
N 1s
Intensity (a.u.)
Binding energy (eV)
Sn 3d
b
Me-4PACz
PhBr-4PACz
200 pA
0 pA
TUNA Current
c
PhBr-4PACz
P=O
O 1s
P-O-H
P=O···Pb
Me-4PACz
Ov/metal oxide
Intensity (a.u.)
Binding energy (eV)
d
Terminal groups
Anchoring groups
Adsorption energy (eV)
Me-4PACz
PhBr-4PACz
e
PhBr-4PACz
Pb 4f
0.30 eV
Me-4PACz
Pb0
Intensity (a.u.)
Binding energy (eV)
f
PhBr-4PACz
I 3d
0.25 eV
Me-4PACz
Intensity (a.u.)
Binding energy (eV)
g
w/VI Int.
w/PbI Int.
Binding energy (eV)
Terminal groups
Anchoring groups
h
Me-4PACz
PhBr-4PACz
tDOS (cm-3 eV-1)
Eω (eV)
i
Evac
4.73
4.88
CB
EF
0.56
0.85
0.44
VB
FTO/Me-4PACz
Perovskite (Me-4PACz)
j
Evac
5.18
5.09
CB
EF
0.48
0.11
0.69
VB
FTO/PhBr-4PACz
Perovskite (PhBr-4PACz)
k
QFLS
ΔVnon-rad
QFLS (eV)
ΔVnon-rad (eV)
Me-4PACz
PhBr-4PACz
l
Me-4PACz D=0.07 cm2s-1
PhBr-4PACz D=0.13 cm2s-1
σx (μm)
Time (ns)
m
Me-4PACz
0.4 ps
0.5 ps
0.6 ps
0.7 ps
0.9 ps
1.3 ps
2.0 ps
3.0 ps
-ΔA (Norm.)
Photon Energy (eV)
n
PhBr-4PACz
0.4 ps
0.5 ps
0.6 ps
0.7 ps
0.9 ps
1.3 ps
2.0 ps
3.0 ps
-ΔA (Norm.)
Photon Energy (eV)
o
Me-4PACz
D=1.76±0.03 cm2/s
S=1569±204 cm/s
1.65 eV
2.07 eV
2.58 eV
Fit
Normalized ΔR/R (a.u.)
Time delay (ps)
p
PhBr-4PACz
D=2.05±0.03 cm2/s
S=3999±132 cm/s
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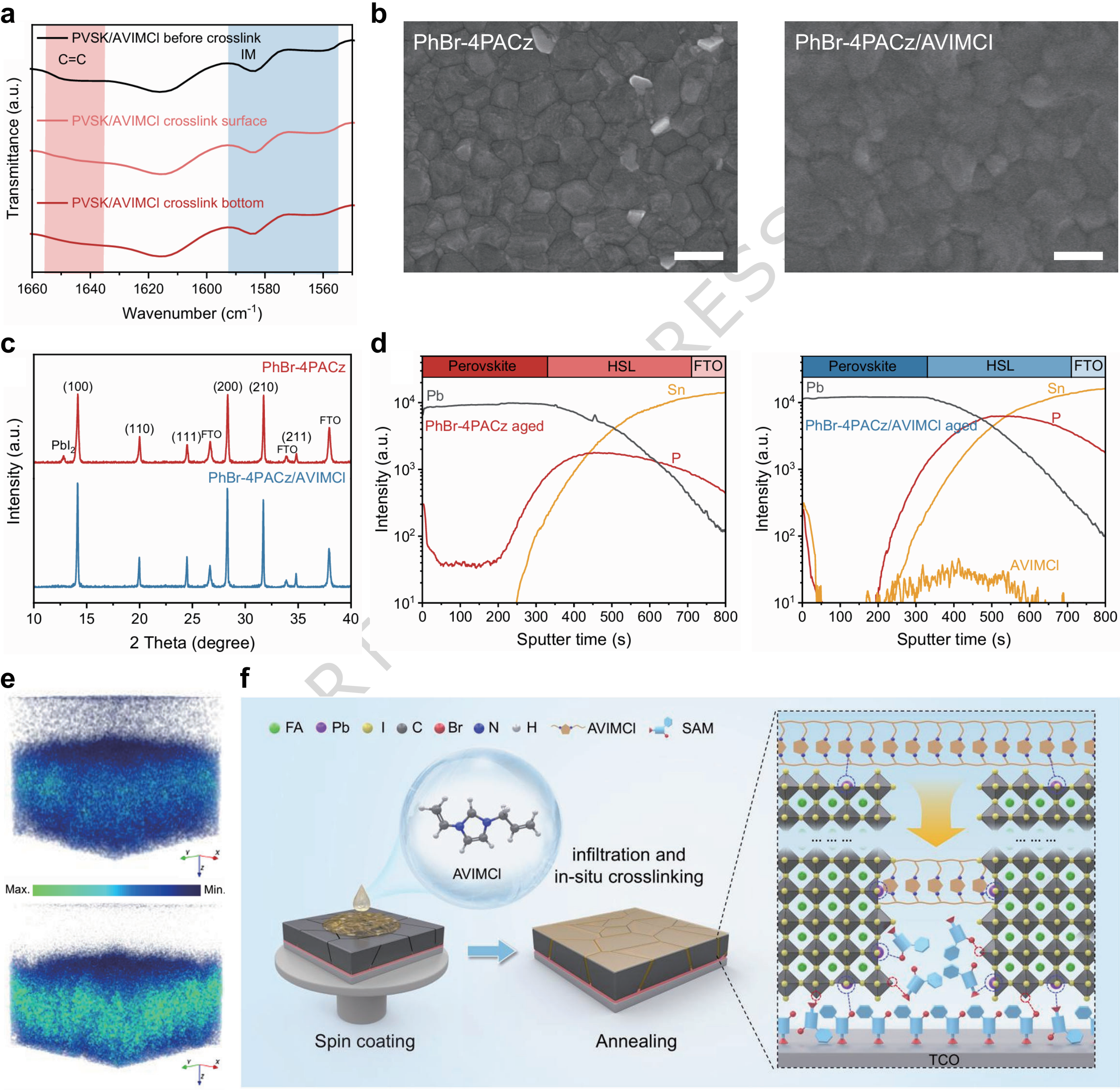
2.07 eV
2.58 eV
Fit
Normalized ΔR/R (a.u.)
Time delay (ps)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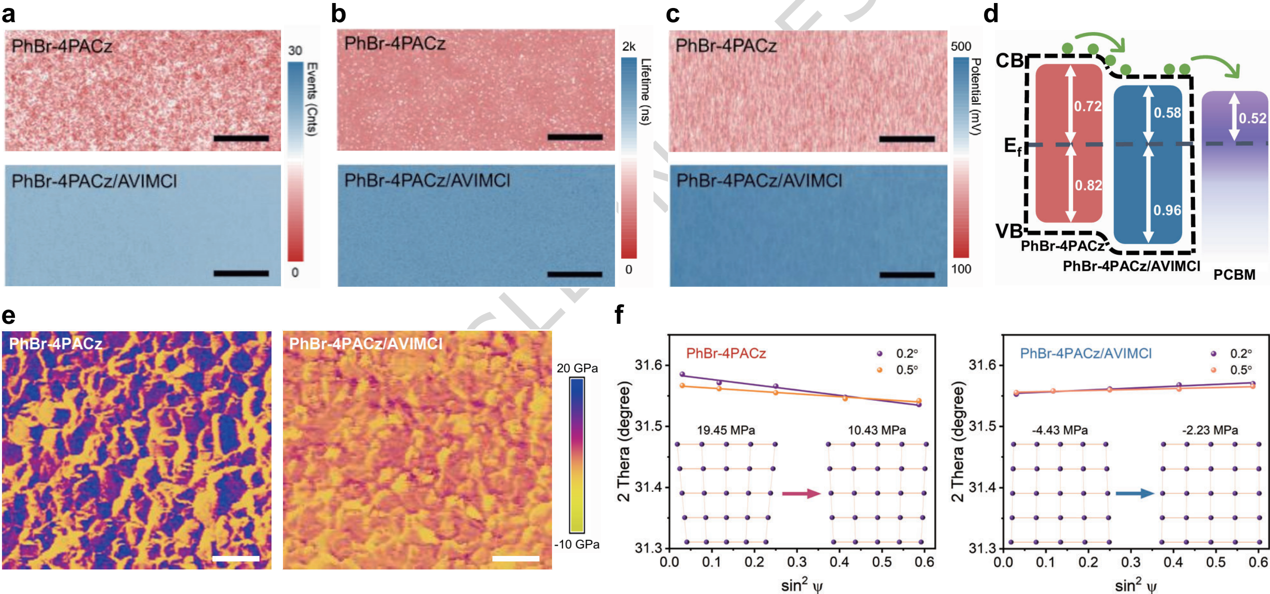
PVSK/AVIMCl before crosslink
C=C
IM
PVSK/AVIMCl crosslink surface
PVSK/AVIMCl crosslink bottom
Transmittance (a.u.)
1660 1640 1620 1600 1580 1560
Wavenumber (cm-1)
b
PhBr-4PACz
PhBr-4PACz/AVIMCl
c
PhBr-4PACz
PbI2
(100)
(110)
(111)
FTO
(200)
(210)
(211)
FTO
FTO
PhBr-4PACz/AVIMCl
Intensity (a.u.)
10 15 20 25 30 35 40
2 Theta (degree)
d
Perovskite
HSL
FTO
Pb
Sn
P
PhBr-4PACz aged
PhBr-4PACz/AVIMCl aged
AVIMCl
Intensity (a.u.)
Sputter time (s)
e
Max.
Min.
f
FA
Pb
I
C
Br
N
H
AVIMCl
SAM
AVIMCl
infiltration and
in-situ crosslinking
Spin coating
Annealing
TCO



a
PhBr-4PACz
PhBr-4PACz/AVIMCl
30
0
Events (Cnts)
b
PhBr-4PACz
PhBr-4PACz/AVIMCl
2k
0
Lifetime (ns)
c
PhBr-4PACz
PhBr-4PACz/AVIMCl
500
100
Potential (mV)
d
CB
Ef
VB
0.72
0.82
0.58
0.96
0.52
PhBr-4PACz
PhBr-4PACz/AVIMCl
PCBM
e
PhBr-4PACz
PhBr-4PACz/AVIMCl
20 GPa
-10 GPa
f
PhBr-4PACz
PhBr-4PACz/AVIMCl
0.2°
0.5°
19.45 MPa
10.43 MPa
-4.43 MPa
-2.23 MPa
2 Thera (degree)
sin2 ψ
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31.5
31.4
31.3
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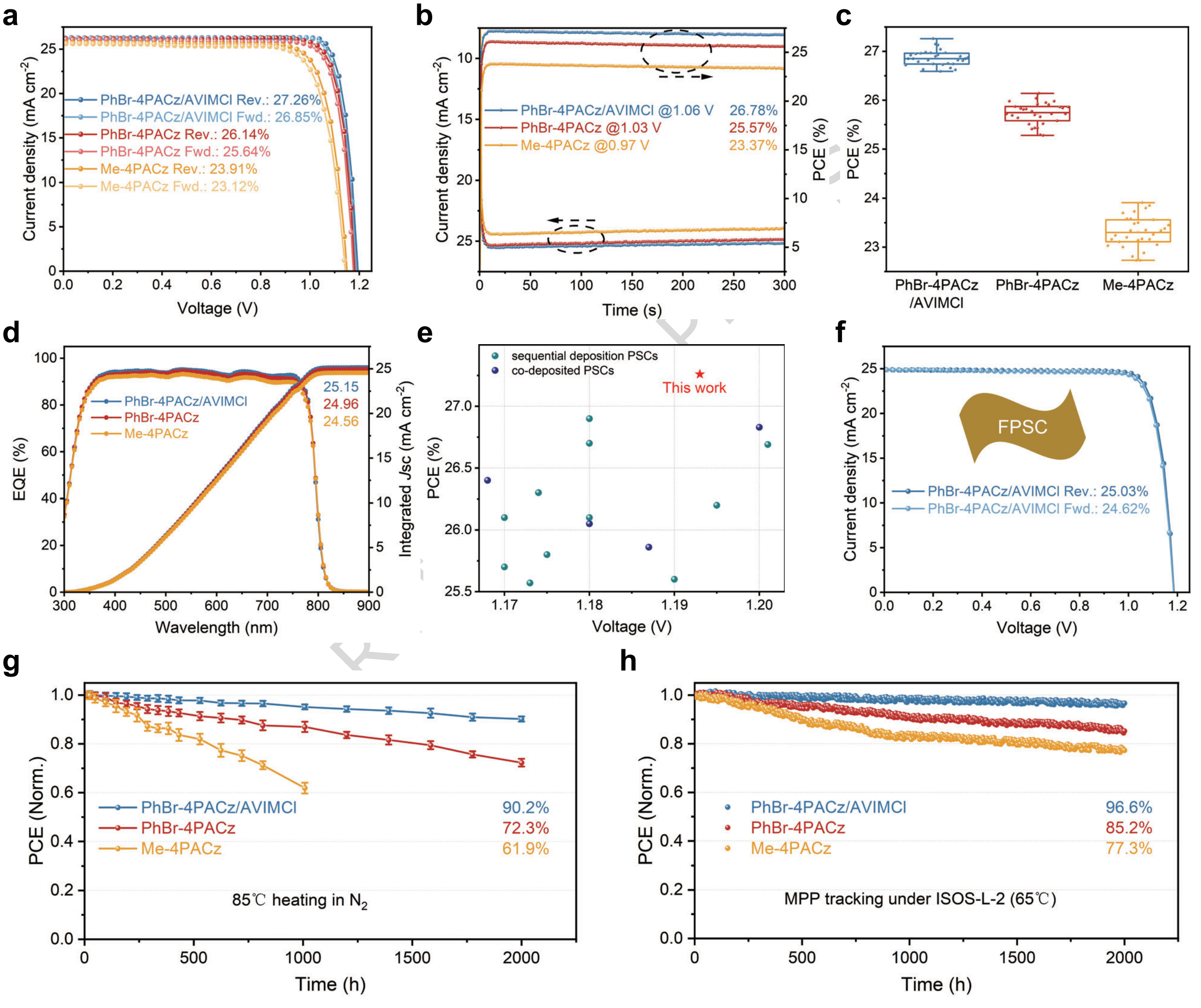
0.1
0.2
0.3
0.4
0.5
0.6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Current density (mA cm-2)
Voltage (V)
PhBr-4PACz/AVIMCl Rev.: 27.26%
PhBr-4PACz/AVIMCl Fwd.: 26.85%
PhBr-4PACz Rev.: 26.14%
PhBr-4PACz Fwd.: 25.64%
Me-4PACz Rev.: 23.91%
Me-4PACz Fwd.: 23.12%
b
Current density (mA cm-2)
PCE (%)
Time (s)
PhBr-4PACz/AVIMCl @1.06 V 26.78%
PhBr-4PACz @1.03 V 25.57%
Me-4PACz @0.97 V 23.37%
c
PCE (%)
PhBr-4PACz /AVIMCl
PhBr-4PACz
Me-4PACz
d
EQE (%)
Integrated Jsc (mA cm-2)
Wavelength (nm)
PhBr-4PACz/AVIMCl
PhBr-4PACz
Me-4PACz
25.15
24.96
24.56
e
sequential deposition PSCs
co-deposited PSCs
This work
PCE (%)
Voltage (V)
f
FPSC
Current density (mA cm-2)
Voltage (V)
PhBr-4PACz/AVIMCl Rev.: 25.03%
PhBr-4PACz/AVIMCl Fwd.: 24.62%
g
PCE (Norm.)
Time (h)
PhBr-4PACz/AVIMCl 90.2%
PhBr-4PACz 72.3%
Me-4PACz 61.9%
85℃ heating in N2
h
PCE (Norm.)
Time (h)
PhBr-4PACz/AVIMCl 96.6%
PhBr-4PACz 85.2%
Me-4PACz 77.3%
MPP tracking under ISOS-L-2 (65℃)